

Deepak Mallampati

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Kent, OH - Open to relocation (Bellevue, WA / Sunnyvale, CA)

PROFESSIONAL SUMMARY

AI/ML engineer (2.5+ years) with production experience building and deploying LLM inference services, agentic AI pipelines, and RAG systems on containerized infrastructure. Delivered measurable outcomes in healthcare AI: ~35% retrieval precision improvement, >30% hallucination reduction, ~30% fewer post-deployment defects in FastAPI/Docker inference services. Strong Python foundation with production API design, ML model serving, and system reliability experience. Seeking to bring LLM application engineering depth to Weights & Biases' Inference AI/ML platform at CoreWeave.

CORE COMPETENCIES

LLM Inference & Serving • Python / FastAPI / Flask • ML Model Deployment • Docker / Containerization • RESTful API Design • Distributed System Patterns • LangChain / LlamaIndex • RAG Systems • LLM Evaluation Pipelines • Agentic Workflows • Observability & Logging • MLOps / LLMOps • scikit-learn / XGBoost / PyTorch • CI/CD (Jenkins)

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare AI

USA

- Packaged ML and **LLM inference as FastAPI/Flask RESTful services**, containerized with Docker, with structured logging, health checks, and load simulation; reduced post-deployment defects ~**30%**.
- Built and iterated **agentic LLM workflows** (LangChain) for multi-step healthcare queries - tool calling, structured outputs, grounding rules; reduced hallucinations >**30%**, improved stability ~**25%**.
- Designed **RAG systems** for clinical knowledge retrieval: semantic chunking, transformer embeddings, vector search; precision improved ~**35%**; retrieval-grounded alignment exceeded **90%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for healthcare classification; improved recall **15-20%** on high-risk categories at precision >90%.
- Maintained HIPAA-conscious data governance: de-identification pipelines, data lineage, audit logging, compliance documentation.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India

- Built **Jenkins CI/CD pipelines** for database deployments; reduced errors >**30%** and improved release cycles ~**35-40%**.
- Designed **Grafana dashboards** for query performance monitoring and incident tracking; cut incident recurrence ~**25%**.
- Optimized T-SQL stored procedures for enterprise ERP system; reduced latency ~**20-25%**.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Nonprofit Tech

Hyderabad, India

- Deployed transformer-based video summarization + document Q&A as **Flask API**; adopted by non-technical staff without training.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent inference pipeline (5 agents) with schema-validated JSON contracts, RAG-grounded CPT/ICD validation, ANN similarity search for duplicate detection, and explainable risk scoring. Greenfield system designed and deployed end-to-end - demonstrates full-stack AI inference ownership.

LangChain, LlamaIndex, Python, FastAPI, Docker, vector search

Manga Lens - Chrome Extension for Real-Time AI Translation

Shipped to Chrome Web Store. Integrates four AI inference providers (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local) via a production extension backend. Multi-provider load routing, 7-day translation cache, per-domain configs for 29 sites.

TypeScript, Manifest V3, Python, multi-provider LLM APIs

Dream Decoder - AI Multimodal Platform

FastAPI + React/TypeScript application with coordinated multimodal API integrations (Perplexity Sonar + Replicate image models). Structured prompt transformation pipeline improved contextual alignment ~30%.

FastAPI, React, TypeScript, multi-modal LLM orchestration

Driver Drowsiness Detection - Real-Time YOLOv8 Fatigue Monitoring

YOLOv8 unified detector replacing two-stage CNN; reduced inference latency ~30%. Sliding-window confidence aggregation reduced false positives ~25%.

YOLOv8, OpenCV, Python, real-time inference

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

TECHNICAL SKILLS

Inference & Serving: FastAPI, Flask, RESTful API design, Docker containerization, model serving patterns, structured outputs, load simulation

LLM & GenAI: LangChain, LlamaIndex, RAG, agentic workflows, LLM evaluation, guardrails, grounding, embeddings, vector search

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, YOLOv8

Languages: Python (primary), TypeScript, React, SQL (PostgreSQL, T-SQL)

Infra & DevOps: Jenkins CI/CD, Grafana, MLOps/LLMOps, observability/logging, cloud-ready deployment

Work Authorization: F-1 OPT - authorized to work in the United States now; future sponsorship required